between being symmetric and antisymmetric. Specifically, eigenfunctions with polynomials of even degree are symmetric, while those with polynomials of odd degree are antisymmetric. The polynomials in this sequence are very well-known. They're called the *Hermite* polynomials. The ground-state eigenfunction $e^{-\frac{\omega}{2}x^2}$, which appears in all of these higher-energy eigenfunctions, is symmetric in *x*.

Fig. 10.3 displays the eigenfunctions for several different energy levels. Each successive eigenfunction oscillates more rapidly than the one before it. This corresponds to an increase in momentum. The more rapidly the wave function oscillates, the greater the momentum of the system. At higher energy levels, the wave function also becomes more spread out. In physical terms, this means the mass is moving farther from the equilibrium point, and moving faster.

These eigenfunctions contain another important lesson. Although they approach zero asymptotically (quite rapidly) they never quite reach zero. That means there is a small but finite chance of finding the particle "outside the bowl" that defines its potential energy function. This phenomenon, known as *quantum tunneling*, is completely unknown in classical physics.

10.8 THE IMPORTANCE OF QUANTIZATION

We've climbed a high mountain in these lectures, but it's not the last mountain. Looking out from the present vantage point, we can get a glimpse of the enormous landscape of quantum field theory. That's material for another book. Or maybe three. But still, we can see a bit of the terrain from where we are.

Consider the example of electromagnetic radiation in a cavity, as shown in Fig. 10.4. In this context, a cavity is a region of space bracketed by a pair of perfectly reflecting mirrors that keep the radiation bouncing endlessly back and forth. Think of the cavity as a long metallic tube that the radiation can travel along in both directions.

There are many wavelengths that can fit into the cavity. Let's consider waves of length λ. Like all waves, these waves oscillate, very much like a